Anikait Kaushik

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Objective

Mechatronics undergraduate open to roles in **Robotics R&D, Mechatronics, and Mechanical Engineering**. Strongest in **mechanical CAD and design** , **robot kinematics and motion control** , **biomechanics and human-motion data analysis**, and **manufacturing & fabrication** – demonstrated through award-winning rover and exoskeleton-rehabilitation systems built end-to-end, from CAD and simulation through fabrication and field testing.

Education

Thapar Institute of Engineering and Technology — **Aug 2023 – May 2027**
B.E. in Mechatronics, Automation and Robotics Engineering; ***CGPA: 9.29/10.0*** *through 6th Semester* — *Patiala, Punjab*

GMSSS MHC, Sector 13 — **85.4%**
Class XII — *Chandigarh*

Jainendra Public School — **96.4%**
Class X — *Panchkula*

Experience

The University of Queensland (UQ) – TIET-UQ Centre of Excellence — **Jun 2025 – Present**
Research Intern – R3: Exoskeleton-enabled Rehabilitation — *Patiala, Punjab*

Mentors: Dr. Ashish Singla, Dr. Sachin Kansal, Alejandro Melendez-Calderon(UQ)

- Contributing to **R3 (Realize, Revive & Recover) – AI Enabled Data-Driven Solutions for Exoskeleton-enabled Rehabilitation**, addressing the need for **data-driven lower-limb rehabilitation** through human-exoskeleton interaction and gait analysis.
- Developed an experimental workflow spanning **exoskeleton CAD and mechanical development, human-motion capture, force measurement, and muscle-activity sensing**, using SolidWorks, FEA, Qualisys, AMTI force plates, and Noraxon EMG.
- Analyzed experimental gait data to evaluate **lower-limb kinematics, gait cycles, Center of Pressure (COP), and Ground Reaction Forces (GRF)**, supporting iterative evaluation of the exoskeleton system.

Orangewood Labs — **Jun 2026**
Robotics & Vision Systems Intern — *Patiala, Punjab*

- Completed the **Robotics & Vision Systems Training & Internship Program** by TIET and Orangewood Labs with hands-on exposure to industrial robotic manipulation and robot programming.
- Programmed an Orangewood robotic arm for **pick-and-place operations** using the **OWL GUI**, executing and testing a series of robot programs.
- Worked with **RoboGPT** for software-assisted robot programming and interaction; received a **Certificate of Achievement** for contribution and dedication during the internship.

Experiential Learning Centre (ELC), TIET — **Jun 2024 – Jul 2024**
Summer Intern – MARVIN Autonomous Cleaning Robot — *Patiala, Punjab*

Mentors: Dr. Ashish Singla, Dr. R.K. Dwivedi

- Identified the need for an **autonomous indoor cleaning system** capable of reducing manual cleaning effort while navigating around obstacles in indoor environments.
- Followed a complete development workflow from **problem identification and concept development to mechanical CAD, electronics integration, fabrication, assembly, and prototype testing**.
- Developed **MARVIN**, integrating a vacuum-based cleaning mechanism, obstacle avoidance, Arduino-based control, and a fabricated mechanical structure using **SolidWorks, 3D printing, and sheet-metal fabrication**; recognized among the **Top 10 ELC Summer Internship Projects**.

Projects

CLIMB CART – Smart Stair-Climbing Load Carrier — **2026 – Present**
Engineering Major Project | Mentors: Dr. Ashish Singla, Dr. Dheeraj Gupta

- Identified a **market gap** through product and user-environment surveys, finding existing stair-climbing carriers to be costly and insufficiently adaptable to narrow Indian staircases, terrain variation, and maintenance requirements.
- Followed a structured **product-development workflow** from market/site survey and requirement definition through concept generation, preliminary design, CAD modelling, iterative mechanism development, FEA/material selection, prototyping, and testing.

Mars Rover – Ganga — **Mar 2025 – Feb 2026**
International Rover Challenge 2026 | Mentors: Dr. Ashish Singla, Dr. Sachin Kansal

- Led mechanical and astrobiology-expedition development of the **IRC 2026 Champion** rover, designing the ABEx multi-axis soil-sampling subsystem integrating excavation, sample handling, sensing, and reagent delivery.
- Redesigned the horizontal sample-handling mechanism around a **single lead-screw/hollow-shaft NEMA-23 stepper**, eliminating dual-actuator synchronization; achieved **150 mm soil penetration** and **12 g sample recovery** across development trials.

Mars Rover – Alaknanda — **Mar 2024 – Mar 2025**
International Rover Challenge 2025 | Mentors: Dr. Ashish Singla, Dr. Sachin Kansal

- Designed and fabricated the complete **soil-collection and testing subsystem**, covering soil extraction, sample handling, onboard chemical testing, and sensor integration.
- Developed and validated mechanical components using **SolidWorks and FEA**, followed by fabrication through 3D printing and CNC machining.

Robotic Kinematic Validation **2026 – Present**
Mentors: Dr. Ashish Singla

- Developed and validated the **LoCoBot–WidowX 200 robotic manipulator** in MATLAB, importing the robot model and validating **forward and inverse kinematics** against the physical robot configuration.
- Developing a **digital twin of the LoCoBot–WidowX 200** in collaboration with **MathWorks**, with the objective of integrating the MATLAB model with the physical robot for real-world motion and validation.
- Gained hands-on exposure to robotic-system installation, commissioning, and verification through interaction with the **Unitree Go2 Pro, Addverb Synchro 5, Alice Service Robot, KUKA KR10 R1420, and Dobot Magician** platforms.

Publications & Patents

- **Published Patent Application (2025):** *An Apparatus, System and Method for Robotic Soil Analysis and Collection*; Named Inventor, Application No. 202511062827.
- **IPRoMM 2026 – Submitted:** *Design and Development of a Multi-Axis Robotic Subsystem for In-Situ Soil Sampling and Analysis.*
- **IEEE Transactions on Robotics – Submitted:** *Experimental Validation of a Unified Euler–Lagrange Framework Using a Three-Link Leg Model for Human Gait Analysis.*
- **Journal of Biomechanics – Manuscript in Final Preparation:** *Load-Dependent Changes in Lower-Limb Joint Kinetics and Hip–Knee Coordination During Walking.*

Leadership

Mechatronics and Robotics Society (MARS), TIET **Aug 2025 – Aug 2026**
General Secretary & Team Captain – Team MARS *Patiala, Punjab*

- Lead and mentor **100+ students** across robotics, mechatronics, mechanical design, and research projects while managing an approximately **Rs. 8,00,000** annual budget.
- Direct multidisciplinary teams for the **International Rover Challenge, International Space Drone Challenge, and DD Robocon**, coordinating technical development, manufacturing, documentation, logistics, and competition operations.
- Served as **Team Lead** for **MARVIN Gen 2, Reconfigurable Robot Gen 1, and Dobot Magician Robot Validation Gen 1**, overseeing project workflows, development-pipeline planning, timelines, finances, documentation, and cross-functional execution.

Technical Skills

Mechanical Design & CAD: SolidWorks, PTC Creo Parametric, AutoCAD, Mechanical Design, CAD Modelling, Product Development, Prototyping
Robotics, Kinematics & Motion Control: MATLAB, Simulink, Robotics System Toolbox, Forward/Inverse Kinematics, Robot Dynamics, Trajectory Planning, Sensor Integration, CubeMars Actuators, ROS 2 (Fundamentals)
Biomechanics & Experimental Systems: Qualisys Track Manager (QTM), AMTI Force Plates, Noraxon EMG, Gait Analysis, Human Motion Analysis, COP, GRF
Manufacturing & Fabrication: Additive Manufacturing, 3D Printing, CNC Machining, Sheet-Metal Fabrication, PCB Fabrication, Ultimaker Cura, Creality Slicer, Bambu Studio
Industrial Automation (Coursework): Pneumatics, Electropneumatics, PLC Fundamentals, RSLogix 500, Digital Electronics – circuit design & simulation using Festo FluidSIM
Simulation & Analysis: *ANSYS, FEA, Simscape, Kinematic Analysis, Human Motion Dynamics (project-level exposure)*
Programming & Software Tools: Arduino IDE, EAGLE, GitHub, Visual Studio Code, Festo FluidSIM, RoboGPT; *Python, C, C++ (basic working knowledge)*
Productivity: Microsoft Excel, Microsoft PowerPoint, Overleaf Latex, Canva

Achievements

- **Champion** – International Rover Challenge 2026, SPROS, MIT Manipal.
- **1st Runner-Up** – International Space Drone Challenge 2026, SPROS, MIT Manipal.
- **1st Runner-Up** – International Rover Challenge 2025, SPROS, Goa.
- **Department Topper** – 10.0 CGPA, Mechatronics, Automation & Robotics Engineering.
- **Certificate of Achievement** – Orangewood Labs, for outstanding contribution and dedication during the internship.
- **Merit Scholar (1st – 4th Year)** – Thapar Institute of Engineering and Technology; Rs. 75,450 / Rs. 1,41,000 / Rs. 1,79,400 / Rs. 1,53,300 respectively.

Certifications & Workshops

- **Robotics & Vision Systems Training & Internship Program** – 2 Weeks, Orangewood Labs & Thapar Institute of Engineering and Technology, 2026.
- **Dynamics of Robotic Systems Workshop** – AIR 2025, IIT Jodhpur; **Practicing Robotics: Reconfigurable Robotics (ROS)** – IIT Ropar, 2025 (1 Day each).
- **MATLAB Onramp** – Self-Paced, MathWorks Training Services, 2025.